Coding & Solution

Date:	29 june 2026
Team ID:	*(x x xx)*
Project Name: Learning	Human Development Index (HDI) Prediction Using Machine
Maximum Marks:	5 Marks

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The **Human Development Index (HDI) Prediction System** is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript** in **Visual Studio Code (VS Code)**. The project predicts the Human Development Index category of a country using four key indicators: **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) Per Capita**.

The application provides a simple web interface where users can enter these values and instantly receive the predicted HDI category (**Very High, High, Medium, or Low**). The machine learning model is trained using historical HDI data and integrated with a Flask web application.

Solution Summary

Field	Details
Repository Link / URL	*Paste your GitHub Repository Link here*
Programming Language(s)	Python 3.x, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy

Key Features Implemented	• HDI prediction using Machine Learning.• Data preprocessing and feature selection.• Model training and evaluation.• Flask-based web application.• User-friendly interface for entering HDI indicators.• Prediction of Very High, High, Medium, and Low HDI categories.• Model saved using Joblib (model.pkl).

Field	**Details**
Pending / Incomplete Features	• Cloud deployment (Render/Heroku).• User authentication and login.• Downloadable prediction reports.• Country comparison dashboard.
Setup / Run Instructions	1. Install Python 3.x and VS Code.2. Install required packages using pip install -r requirements.txt.3. Run train_model.py to train the model (if needed).4. Run python app.py.5. Open http://127.0.0.1:5000 in a web browser.6. Enter the required indicators and click **Predict** to view the HDI category.

Code Quality Checklist

S.No	**Criteria**	**Status**
1	Code is modular and organized into functions/classes.	**Yes**
2	Meaningful variable and function names are used.	**Yes**
3	Code includes comments/documentation where necessary.	**Yes**
4	Error handling is implemented for critical operations.	**Yes**
5	The application runs without critical errors.	**Yes**
6	Code is committed to a version control repository (GitHub).	**Yes**

Additional Notes / Comments

- The project is developed using **Visual Studio Code** following Python coding standards (PEP 8).
- The source code is divided into separate folders (dataset, models, templates, static) for better organization.
- The machine learning model is trained using **Scikit-learn** and stored as model.pkl using **Joblib**.
- The Flask framework connects the frontend with the trained model to provide real-time HDI predictions.
- The project is maintained in a **GitHub repository**, enabling collaboration among team members and version control.